Five-Dimensional Reliability Evaluation for Enterprise AI Agents: A Reproducible Framework and Synthetic Stress-Test Suite

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Abstract

Agentic artificial intelligence systems execute multi-step trajectories involving reasoning, tool use, memory, and external state changes. Mean task success can obscure operational properties required for deployment. We introduce a five-dimensional evaluation framework covering task fidelity, recovery robustness, action safety, cost-latency efficiency, and evidence-observability. We release a v0.2 reference implementation, synthetic enterprise stress-test tasks, demonstration outputs, adversarial unit tests, and a reproducible notebook. The framework uses geometric aggregation while treating unobserved recovery as missing rather than perfect. The release is explicitly methodological: synthetic results illustrate scoring behavior and are not claims about real language models. We specify a protocol for extending the suite to real agent systems through repeated trials, controlled perturbations, policy-boundary tests, and trace auditing.

1 Introduction

LLM agents increasingly interleave reasoning with action [**?**], use feedback and memory [**?**], and operate in interactive environments. AgentBench [**?**], WebArena [**?**], GAIA [**?**], WorkArena [**?**], and OSWorld [**?**] illustrate this shift. Task success alone does not capture consistency, recovery, authorization boundaries, economics, or auditability.

2 Framework

For each run we compute five normalized dimensions: Task Fidelity F, Recovery Robustness R, Action Safety S, Cost-Latency Efficiency C, and Evidence-Observability E. For observed dimensions with normalized weights w_i, we define

$$Q = \exp\left(\sum_i w_i \log(\max(\epsilon, x_i))\right).$$

If a dimension is genuinely unobserved (for example no recovery event was tested), v0.2 does not award a perfect value; the default composite renormalizes across observed dimensions and reports the missing component separately.

3 Synthetic Stress-Test Suite

The release contains enterprise workflow task specifications paired with perturbations such as tool timeout, malformed output, prompt injection, stale state, and permission boundaries. All bundled demonstration observations are labeled synthetic.

4 Reference Implementation and Correctness

Version 0.2 validates finite inputs, enforces metric ranges, rejects impossible recovery counts, distinguishes unobserved recovery from successful recovery, and adds tests for these edge cases. A helper computes repeated-run consistency from binary outcomes; real studies should derive consistency from repeated executions rather than manually assigning it.

5 Protocol for Real-Agent Validation

Future empirical work should preregister agent versions, prompts, tools, budgets, success conditions, perturbations, and primary metrics; execute repeated trials; report uncertainty and effect sizes; retain raw trajectories; and avoid using the synthetic demonstration numbers for comparative model claims.

6 Limitations

Weights are domain dependent and several submetrics require task-specific operationalization. The present release is a reproducible measurement framework, not an empirical ranking of agents.

7 Conclusion

Reliable enterprise agents should be evaluated on more than whether they eventually reach a goal. The framework makes five operational dimensions explicit and supplies transparent software for testing and extending the proposal.